Mohtashim Ul Haq

Software Engineer | .NET, Blazor and AI Systems | Fintech and Healthcare
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PROFESSIONAL SUMMARY

Software engineer with over five years of experience building secure, production-grade web systems in fintech and healthcare on the .NET stack. Led the development of an AI-assisted cancer diagnostic platform for whole-slide images in partnership with 3 major hospitals; enhanced a card payment gateway and integrated multiple banking processors; and recently designed and built, single-handedly, IronForge, a multi-tenant SaaS platform with governed AI agents (about 54,000 lines of C#, about 300 automated tests, CI/CD with automatic rollback). Now pursuing a Master of Data Science at Universiti Sains Malaysia, focused on AI for digital pathology.

TECHNICAL SKILLS

- **Languages:** C#, Python, SQL, JavaScript, Razor, HTML, CSS, Bash
- **Backend:** .NET 10, .NET 6, .NET Framework, ASP.NET Core Web API, ASP.NET MVC, Entity Framework Core, REST and SOAP web services, background jobs
- **Frontend and mobile:** Blazor (Server and Hybrid), .NET MAUI (Android, iOS, macOS, Windows), Angular, jQuery, Bootstrap
- **AI and data:** AI agents with tool calling, OpenAI via Microsoft.Extensions.AI, Model Context Protocol (MCP), Python, PyTorch, whole-slide image processing
- **Databases:** PostgreSQL, SQL Server, MariaDB
- **Architecture:** Clean architecture, multi-tenant SaaS, transactional outbox, idempotent processing, optimistic concurrency
- **Security:** JWT with rotating refresh tokens, TOTP two-factor authentication, role- and permission-based authorization, rate limiting, audit trails
- **DevOps and testing:** Docker, GitHub Actions CI/CD, Azure DevOps, Caddy, IIS, Apache, Linux, OpenTelemetry, xUnit, Moq, Testcontainers
- **Integrations:** Stripe Connect, banking processors, ZKTeco fingerprint terminals, SMTP (MailKit), Cantaloupe image server

PROFESSIONAL EXPERIENCE

Software Engineer | CROEM Inc., Islamabad, Pakistan **Oct 2021 – Jul 2026**

Neo Gateway (payment gateway)

- Enhanced a **payment gateway** for efficient card-present (CP) and card-not-present (CNP) transactions.
- Integrated multiple banking processors into the gateway, increasing daily transaction volume and contributing to substantial revenue growth.
- Optimized backend performance, resulting in faster transaction processing.

Xpress Pago (subscription platform)

- Developed a web portal for managing customer **subscriptions and payment processes**.
- Implemented reporting and analytics that improved business decision-making and operational efficiency.

Technologies: .NET Framework, C#, ASP.NET MVC, jQuery, SQL Server, Windows and web services, IIS, Azure DevOps

Software Engineer | InfoMatrix, Islamabad, Pakistan **Nov 2020 – Sep 2021**

Amal (AI-based cancer diagnostic system for digital pathology)

- Led the development of Amal **from scratch**: a web platform that lets pathologists and researchers view, annotate and analyse gigapixel whole-slide images (WSIs) of tissue for cancer diagnosis.
- Built processing pipelines for large-scale medical images and served them as tiles through the Cantaloupe image server, so slides open and zoom smoothly in the browser.
- Built annotation workflows for pathologists and integrated PyTorch deep learning models for cancer diagnosis into .NET 6 backend services.
- Secured partnerships with **3 major hospitals and 1 university**, obtained initial funding, and established the system's credibility in the medical field.

Technologies: Python, PyTorch, LibTIFF, .NET 6, jQuery, MariaDB, Cantaloupe Image Server, Docker, Git, Apache, Linux

PROJECTS

IronForge | Multi-tenant gym management SaaS with governed AI agents **2026**

Solo developer: product design, architecture, backend, frontend, mobile, AI and DevOps

Live: app.mohtashim.dev **|** **Source code:** github.com/Mohtashim890/IronForge

- Designed and built a production-grade SaaS with four connected apps: a Blazor web dashboard for gym staff, a .NET MAUI member app for Android, iOS, macOS and Windows, a platform console for the operator, and a REST API with an MCP server. Feature-complete v1, deployed by CI/CD.
- Built 3 AI agents with 24 governed tools (OpenAI via Microsoft.Extensions.AI) that act only through permission-checked business services, with risk-based human and four-eyes approvals, amount caps, monthly token quotas and a full audit trail; exposed the same tools to external agents over the Model Context Protocol.
- Implemented row-level multi-tenancy with EF Core global query filters on PostgreSQL, secure by default and proven by integration tests; built an idempotent billing engine (PostgreSQL advisory locks, transactional outbox, idempotent Stripe Connect webhooks, optimistic concurrency).
- Implemented custom JWT authentication with rotating refresh tokens, TOTP two-factor authentication and rate limiting; designed HMAC-signed offline QR passes; integrated ZKTeco fingerprint terminals over their push protocol so they refuse unpaid members even offline, without storing biometric data.
- **Evidence:** about 54,000 lines of C# across 15 projects, about 300 automated tests (xUnit, Testcontainers on real PostgreSQL), 33 database migrations; GitHub Actions CI/CD to Docker on a VPS with pre-migration backups, health checks, automatic rollback and OpenTelemetry monitoring.

AI Pathology Platform | Master's industry project, USM **2026 – present**

In design and early development; validation planned with a partner pathology laboratory

- Designing an end-to-end platform that checks slide quality, runs rotation-equivariant deep learning models with calibrated uncertainty, and shows results to pathologists in a Blazor and OpenSeadragon viewer with a triage worklist, similar-case search and structured HL7 FHIR reports.
- Architecture: event-driven .NET services (ASP.NET Core, RabbitMQ workers, ONNX Runtime inference), PostgreSQL with pgvector, object storage, and Python with PyTorch for model training; planned cloud deployment on Azure with Terraform and GitHub Actions.

EDUCATION

Master of Data Science | Universiti Sains Malaysia (USM), Malaysia **2026 – expected 2028**

- Coursework: Machine Learning; Big Data Storage and Management; Enabling Technologies and Infrastructures for Big Data; Data Visualisation and Visual Analytics; Multimodal Information Retrieval; Text and Speech Analytics; Business Intelligence and Decision Analytics.

Bachelor of Engineering in Computer Engineering | CEME, NUST, Islamabad, Pakistan **2013 – 2017**

- Coursework: Artificial Intelligence and Machine Learning, Data Structures, Software Engineering, Digital Signal Processing, Computer and Mobile Networks. Awarded a GPA scholarship.

PUBLICATION

Sana Hanif et al. *A Validation Study: Demonstrating Utility of Virtual Microscopy in Comparison to Glass Microscopy in Gynecologic Pathology*. Manuscript under review.

LANGUAGES AND INTERESTS

Languages: English (fluent), Urdu (native). **Interests:** reading in science, religions, philosophy and history.